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CS502 QUIZ 1 (LEC 19 to 28





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Which method is preferable for dealing with chain matrix multiplication?

i. Divide and conquer strategy
ii. Dynamic programming formulation
iii. Graph theory
iv. Greedy Approach

Huffman algorithm produces the…….prefix code tree.

i. Better
ii. Optimal
iii. Worst
iv. Best

A….w is adjacent to vertex v if there is an edge from v to w.

i. Acyclic
ii. Vertex
iii. Loop
iv. Cycle

Using ASCII standard the string "greedy" will be encoded with

i. 44 bits
ii. 120 bits
iii. 40 bits
iv. 48 bits

Find the maximum value of the items which can carry using knapsack weight capacity =50

Item weight	**Value**
10	**70**
20	**20**
30	**80**
70	**200**

i. 90
ii. 280
iii. 200
iv. 100

In activity scheduling algorithm, each activity is represented by a

i. Rectangle
ii. Square
iii. Circle
iv. Triangle

Those problems in which greedy finds good, but not always best is called a greedy…..

i. Heuristic
ii. Solution
iii. Result

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iv. Algorithm

The knapsack problem belongs to the domain of…..Problems

i. Searching
ii. Sorting
iii. Linear solution
iv. Optimization

The general coin change problem can be solved using

i. Recursion
ii. Greedy algorithm
iii. Dynamic programming
iv. Divide and conquer

Huffman algorithm produces the……prefix code tree

i. Optimal
ii. Better
iii. Best
iv. Worst

Huffman algorithm generates an optimum …..code

i. Postfix
ii. Infix
iii. None of the given options
iv. Prefix

There are …..ways of representing graphics

i. 2
ii. 1
iii. 3
iv. 4

Knapsack word originates from ……language

i. German
ii. English
iii. French
iv. Norwegian

Graphs are important…….model for many application problems

i. Mathematical
ii. Unpredictable
iii. Haphazard
iv. Unsystematic

Which type of algorithm is harder to prove the correctness?

i. Dynamic
ii. Greedy

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iii. Divide and conquer
iv. Brute force

…..Items are not allowed in 0/1 knapsack problem

i. Fractional
ii. 0
iii. 1
iv. 0/1

Matrix multiplication is a(n)……..operation

i. Nether commutative nor associative
ii. Transitive
iii. Commutative
iv. Associative

For a Diagraph G= (V.E), Sum of in-degree (v) ----------.

- Not equal to sum of out-degree(v)
- = sum of out-degree(v) pg#115
- < sum of out-degree(v)
- > sum of out-degree(v)

DFS or BFS yields a----------of the graph.

- Traversed tree
- Spanning tree pg#125
- Simple tree
- Free tree

Using ASCII code, each character is represented by a fixed-length code of----------bits per character.

- 4
- 6
- 8 pg#100
- 10

In Knapsack Problem, the goal is to put items in the Knapsack such that the value of the items is---------

-------subject to weight limit of the Knapsack.

- Minimized
- Decreased
- Maximized pg#109
- None of the above given

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Consider the string"abcdaacac",if the string is coded with ASCII codes, the message length would be---

- 70 bits
- 60 bits
- 90 bits
- 72 bits

A graph is said to be acyclic if it contains---------.

- At least one cycle
- Exactly one cycle
- Always more than one cycle
- No cycles pg#116

The number of edges that come out of a vertex is called the----------of that vertex in the digraph.

- Post-degree
- in-degree
- out-degree pg#114
- pre-degree

If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- 3x2
- 2x3
- 2x2
- 3x3pg#87

A/an----------is one in which you want to find, not just a solution, but the best solution.

- Optimization problem
- Divide and Conquer
- NP complete problem
- Best problem

Fractional Knapsack is founded on----------method.

- Greedy
- Recursive
- Divide and Conquer
- Dynamic programming

If the graph is represented using an adjacency matrix, then Breadth-first search takes----------time.

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- O(E+1)
- O(V^2)
- O(V)
- O(E)

In inductive approach of Knapsack problem, we consider 2 cases, ----------or----------.

- Median, Mode
- Recursive, Iterative
- Leave object, Take object pg#93
- Sequentially, Parallel

A Greedy algorithm can NOT be used to solve all the----------problems.

- Dynamic programming (Google)
- Memorization programming
- Edit-distance programming
- Storing value programming

In Huffman encoding, the---------is the number of occurrences of a character divided by the total characters in the message.

- Counting
- Parsing
- Relative Probability pg#100
- Weight

The Binary Tree constructed by a Huffman Encoding is a:

- Full Binary Tree pg#102
- Partial Binary Tree
- Incomplete Binary Tree
- None of the given option

Following is not the application of Edit Distance Problem.

- Speech recognition pg#76
- Spelling correction
- Ascending order
- Computational Molecular Biology

Consider three Matrices X, Y, Z of dimensions 1x2, 2x3, 3x4 respectively. The number of multiplication of(XYZ) is:

- 18
- 32
- 24
- 30

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18) In ---------- Knapsack Problem, limitation is that an item can either be put in the bag or not. Fractional items are not allowed.

- 0
- 1
- 0/1 pg#91
- Fractional

An in-place sorting algorithm is one that----------uses additional array for storage.

- Always
- Permanently
- Does not pg#54
- Sometime

If Matrix-A has dimensions “pxq” and Matrix-B has dimensions “qxr”, then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- P x q
- P x r page#84
- q x r
- q x p

Counting sort is suitable to sort the elements in range 1 to K.

- K is large
- K is small pg#71
- K may be large or small
- None

When matrix A of 5x 3 is multiply with matrix B of 3x 4 then the multiplication required is:

- 15
- 12
- 36
- 60

----------is a linear time sorting algorithm.

- Merge sort
- Quick sort
- Bubble sort
- Radix sort

In Dynamic Programming approach, we do not store the solution to each sub problem in case if it reappears.

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- True
- False page#75

Dynamic Programming approach is usually useful in solving optimization problem.

- True page#97
- False

Which of the following algorithm provides an optimal solution for the activity selection problem?

- Divide and Conquer
- Brute force
- Greedy pg#105
- Recursive

A graph is---------- if every vertex can reach every other vertex.

- Connected pg#116
- Cycle
- Acyclic
- Loop

In a Huffman encoding when a new node is created by combining two nodes, the new node is placed in the _____.

- Priority queue pg#100
- Linked list
- Min heap tree
- Graph traversal

Huffman algorithm produces the _____ prefix code

- Optimal pg#105
- Best
- Worst
- Better

In _____ algorithm, you hope that by choosing a local optimum at each step, you will end up a global optimum.

- Simple
- Divide and conquer
- Greedy pg#97
- Brute Force

The string “Imncde” is coded with ASCII code, the message length would be ____ bits.

- 24
- 36

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- 48
- 60

For graph traversal, breadth-first search strategy _____

- Is always recursive
- Cannot br recursive
- Cannot be non-recursive
- Can be both recursive and non-recursive page 119

In activity scheduling algorithm , the width of a rectangle _____

- Is always ignored
- Directs towards recursion
- Should be maximized
- Indicates the duration of an activity pg#106

If the graph is represented using an adjacency list, then Breadth-first search takes--------- time

- O(V^2)
- O(V)
- O(V+E) pg#138
- O(E+1)

Suppose you are given infinite coins of 1,2 ,3, and 4.Select the ways of the minimum number of coins that required to achieve a sum of 6:

- 1
- 2 conceptual
- 3
- 4

sing ASCII standard the string “greedy” will be encoded with

- 48 bits Conceptual
- 120 bits
- 44 bits
- 40 bits

The Huffman codes provide a method of -------data efficiency.

- Reading/Writing
- Encoding/Decoding pg#99
- Divide/Conquer
- Inserting/Deleting

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In the context of activity selection algorithm, time s dominated by sorting of the activities by------

- Start Times
- Finish Times pg#106
- Average Times
- CPU Burst Times

Time complexity of the “0-1” knapsack algorithm depends on----

- Number of items
- Capacity of the knapsack
- Size of the Table
- Number of items and capacity of knapsack (not sure)

The greedy approach gives us an optimal solution when the coins are all powers of a -------- denomination

- Fixed pg#98
- Variable
- Constant
- Static

In Activity Selection, we say that two activities are non-interfering if their start-finish interval-------

overlap

- Do
- Do not pg#105
- Sometimes
- Once

How many steps are involved to design the dynamic programming strategy?

- 2
- 3
- 1
- 4 pg#92

Bag is a----------

- type of algorithm pg#119
- data structure
- program
- compiler

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The running time of brute-force algorithm to solve Knapsack problem is------

- O(n)
- O(n3)
- O (n!)
- O(2^n) pg#92

If a matrix has three rows and two columns, the dimension of matrix will be:

- 3x2 Conceptual
- 2x3
- 3x3
- 2x2

The Probability in Huffman encoding is the number of occurrences of a character divided by the total-

----------in the message.

- Numbers
- Frequencies
- Strings
- Characters pg#100

In recursive formulation of Knapsack
Problem: V [0, j] = ---------- for j>=0

- -1
- 0 pg#93
- 1
- 2

The Knapsack Problem belongs to the domain of----------problem.

- Optimization pg#91
- Sorting
- Linear solution
- Searching

An optimization problem is one in which you want to find the----------solution.

- Simple
- Good
- Best pg#97
- Worst

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Huffman algorithm generates an optimum---------- code.

- Prefix page 102
- Postfix
- Infix
- None of the above

Which of the following algorithms solves the Fractional Knapsack Problem more effectively?

- Divide and Conquer
- Greedy algorithm page 109
- Dynamic programming
- Backtracking

If we implement the bag by using a queue, we have----------.

- BFS page 124
- DFS
- Graph
- Loop

One of the limitation in 0/1 Knapsack is that an item can either be----------in the bag or not.

- Use
- Put page 91
- Move
- Store

In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object'i' then we gain----------.

- W(Total weight of Knapsack)
- V(Total value of all items)
- vi(value of object i) page 93
- None of the given option

An activity scheduling algorithm, the width of a rectangle----------.

- Is always ignored
- Direct toward recursion
- Should be maximized
- Indicates the duration of an activity page 106

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The prefix code generated by Huffman algorithm---------- the expected length of the encoded string.

- Minimizes page 102
- Balances
- Maximizes
- Keeps constant

In Huffman encoding, the characters with smallest probabilities are placed at the----------depth of the tree.

- Minimum
- Average
- Maximum page 102
- Root

There are ---------- ways to representing graphs.

- 3
- 1
- 2 page 116
- 4

Each time we traverse graph by Breadth-first search algorithm, we count the distance from----------.

- Starting node
- Neighbours of the starting node page 117
- Right most node
- Left most node

In Activity Selection problem, intuitively----------.

- There are always short activities as input
- Short activities are not attractive
- Duration of the activities does not matter
- We do not like long activities page105

62) The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an----------solution.

- Simple
- Sub-optimal
- Optimal page 105
- Non optimal

The ---------- is a problem for which the greedy algorithm approach provides an optimal solution.

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- Activity selection page 105
- Dynamic programming
- Knapsack problem
- NP complete problem

Which of the following ways can be used to represent a graph?

- Adjacency list
- Incidence matrix
- Adjacency list , Adjacency Matrix page 116
- No way to represent

Queue Data Structure work on ---------- principles.

- FIFO(first in first out) cs301
- LIFO(last in first out)
- JLO(just in last out)
- LOFI(last out first in)

Graphs can be represented by an---------- and ----------.

- queue , stack
- adjacency list , adjacency matrix page 116
- adjacency right , adjacency left
- Binary , linear

Identify a TRUE statement about Knapsack.

- The Knapsack Problem does not belongs to the domain of optimization problems
- The Knapsack Problem belongs to the domain of optimization problems page 91
- The Knapsack Problem cannot be solved by using Dynamic programming
- The Knapsack Problem is optimally solved by using Brute force algorithm

Which of the following is true about graph?

- A graph may contain no edges and many vertices
- A graph may contain many edges and no vertices

Which type of algorithm is harder to prove the correctness?

- Greedygoogle
- Brute force
- Dynamic programming
- Divide and Conquer

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